

Semantic Caching: 30x Faster RAG Response Time

Leveraging intelligent query caching to dramatically reduce latency in production AI systems

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Tech Stack

- Ollama Llama 3.2
- Nomic-embed-text
- FAISS vector store
- Semantic cache layer

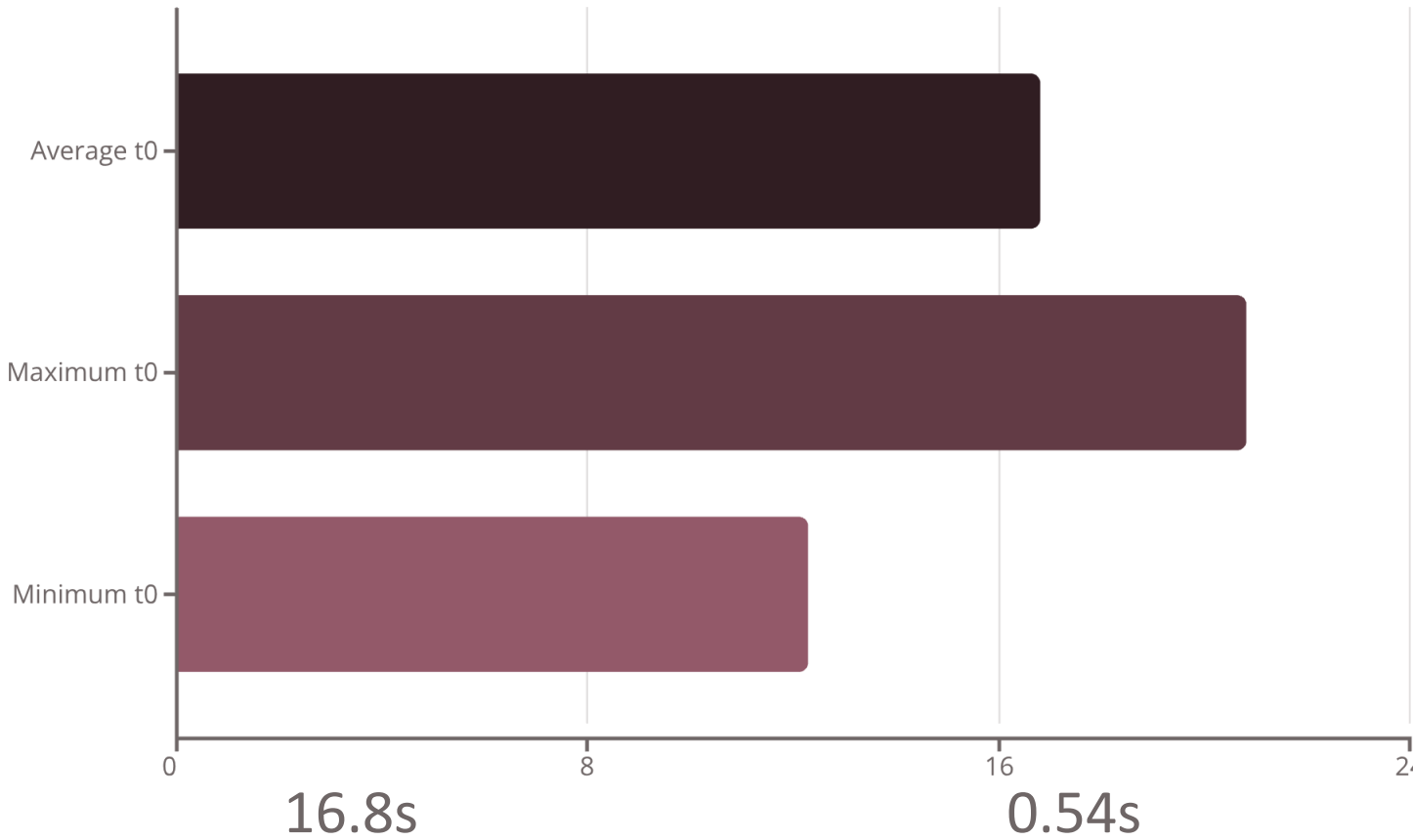
Test Parameters

- 50 Q&A pairs tested
- Real-world queries
- Production conditions
- Consistent methodology

Performance Gains

- 96.7% latency reduction
- Sub-second cached queries
- Consistent improvement
- Scalable architecture

Initial Query Performance



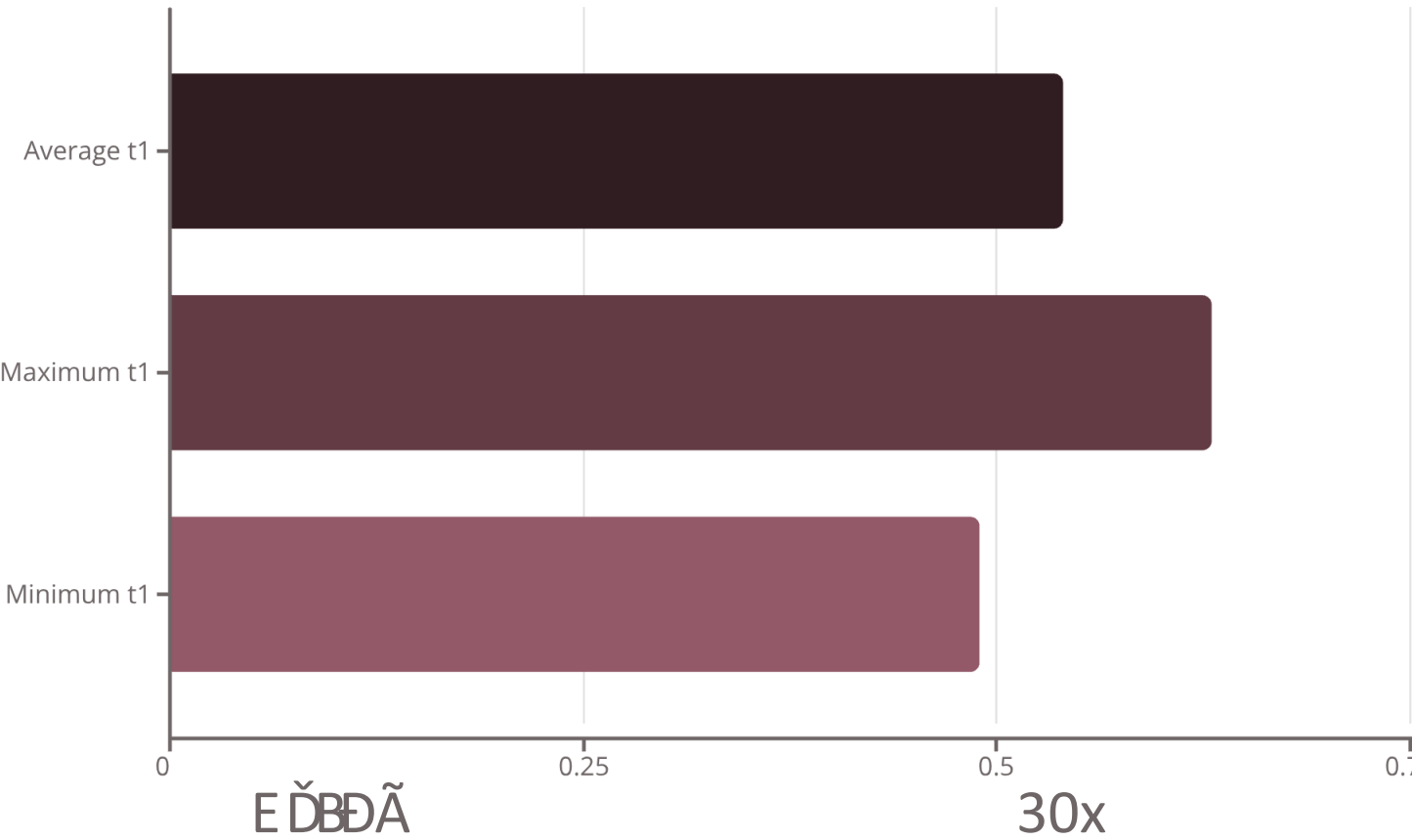
16.8s

0.54s

Average Initial Query

Average Cached Query

Cached Query Performance



0.54s

30x

Latency Reduction

Speed Multiplier

First-time query processing through full RAG pipeline

Semantic cache hit delivers near-instantaneous results

Consistent performance improvement across all test cases

Cached queries execute 30 times faster than initial queries

